

Me-AI

AI-Powered Personalized Meal Recommendation Platform

The concept designed in CENG 407 evolved into a secure, modular, and fully working full-stack product in CENG 408.

80

API routes

REST workflows across rec / inventory / consumption / admin

30

Flyway migrations

Versioned database evolution

260

Backend tests

Unit checks across 47 test files

Berk Memiş · Burak Metin · Yiğit Efe Salgın

Supervisor: Asst. Prof. Abdül Kadir GÖRÜR

Problem & Motivation

Meal planning fails when recipe search, nutrition tracking, inventory and AI live in separate worlds.

Static lists

Generic recipe apps filter catalog items, but rarely adapt to daily context.

No inventory signal

Users decide from what they actually have at home, not abstract meal plans.

Weak personalization

Allergies, diet type, disliked ingredients and calorie goals need different priority.

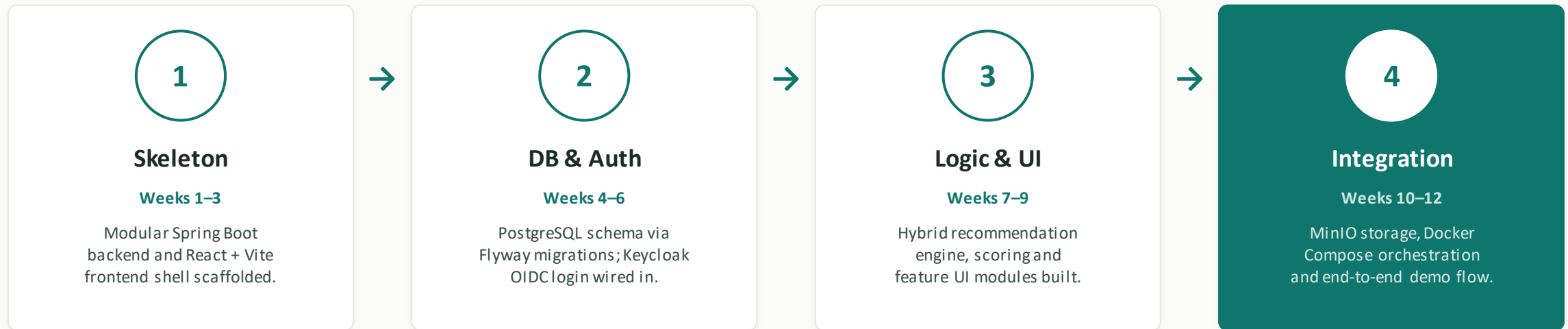
Unchecked AI

LLMs can invent nutrition or ingredients unless grounded in verified data.

Design response: keep safety-critical filtering and nutrition math in backend services; use AI only after the candidate set is constrained.

407 to 408

The project moved from design intent in CENG 407 to an integrated, working full-stack product across CENG 408.



Across roughly twelve weeks, the CENG 407 design intent became a working, containerized full-stack product delivered in CENG 408.

Project Scope

The delivered scope is software engineering for personalized meal planning, not medical nutrition automation.

What it is

A web-based platform combining user profiles, inventory, recipes, nutrition estimates, recommendations and consumption tracking.

What it deliberately avoids

It does not train a medical model or claim clinical-grade diet advice. Nutrition values remain estimates.

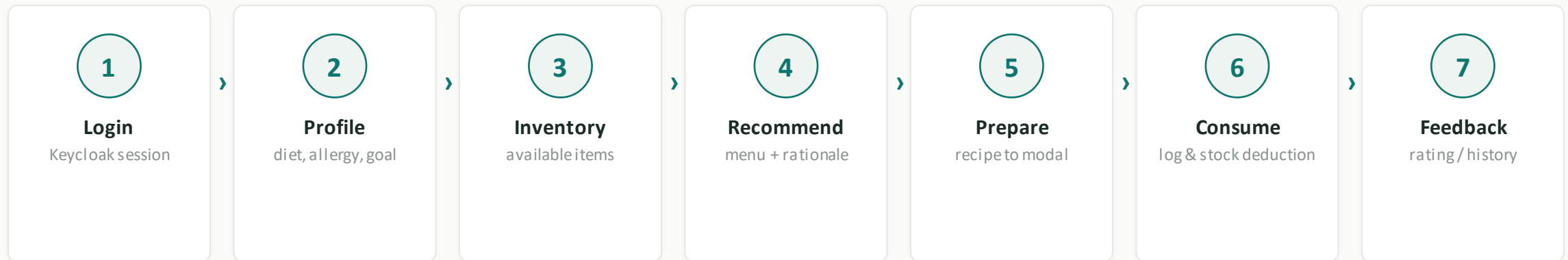
Why that matters

The engineering challenge is controllability: verified data and deterministic safeguards before AI language.

Positioning line for the jury: an applied AI + full-stack architecture project with responsible boundaries.

User Journey

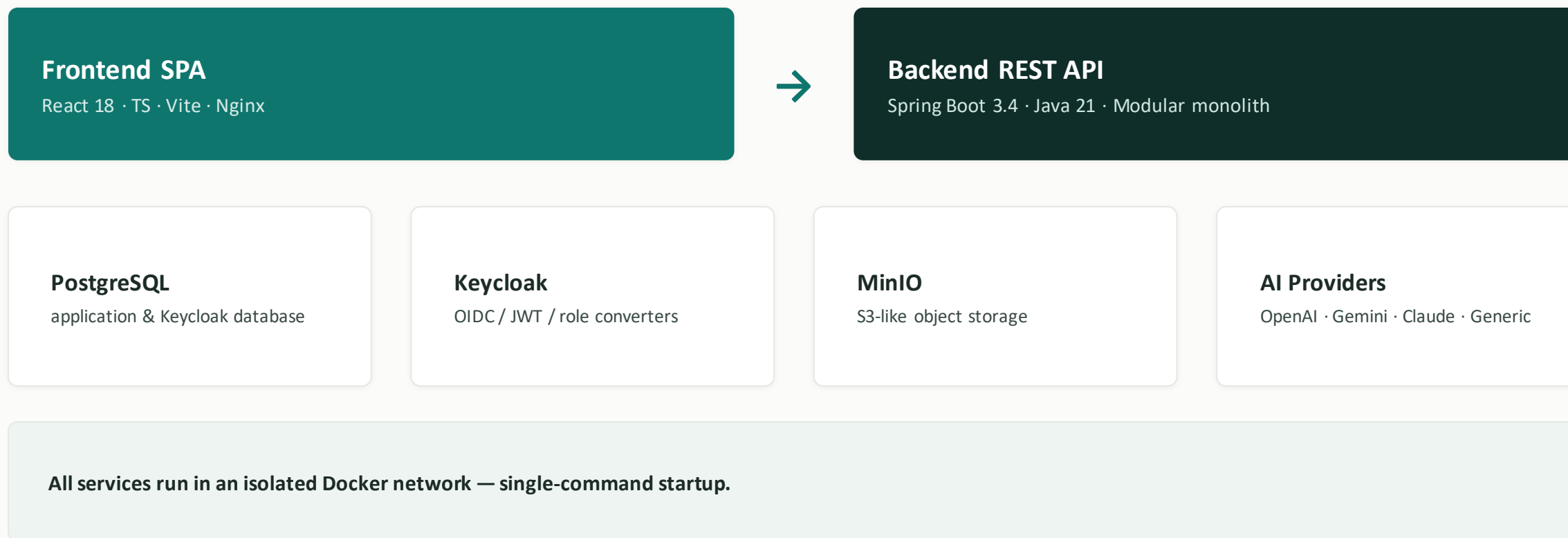
A recommendation becomes useful only when it closes the loop from profile to pantry to actual consumption.



This end-to-end path proves user value and system integration at the same time.

System Architecture

The runtime is a containerized full-stack system, while the backend remains a modular Spring Boot application.



Backend Architecture

Thin client architecture keeps business rules, scoring and unit conversion in one backend source of truth.

01 — Infrastructure Layer External service clients · MinIO storage adapters · Testcontainers helper modules

02 — Domain Layer (the brain) Entities · repositories · custom business services · hybrid recommendation scoring

03 — Application Layer Controllers · DTOs · Keycloak role converters · mappers · REST endpoints

04 — Utilities Layer Dataset processing utilities · unit converters · shared logging tools

Frontend Renders State

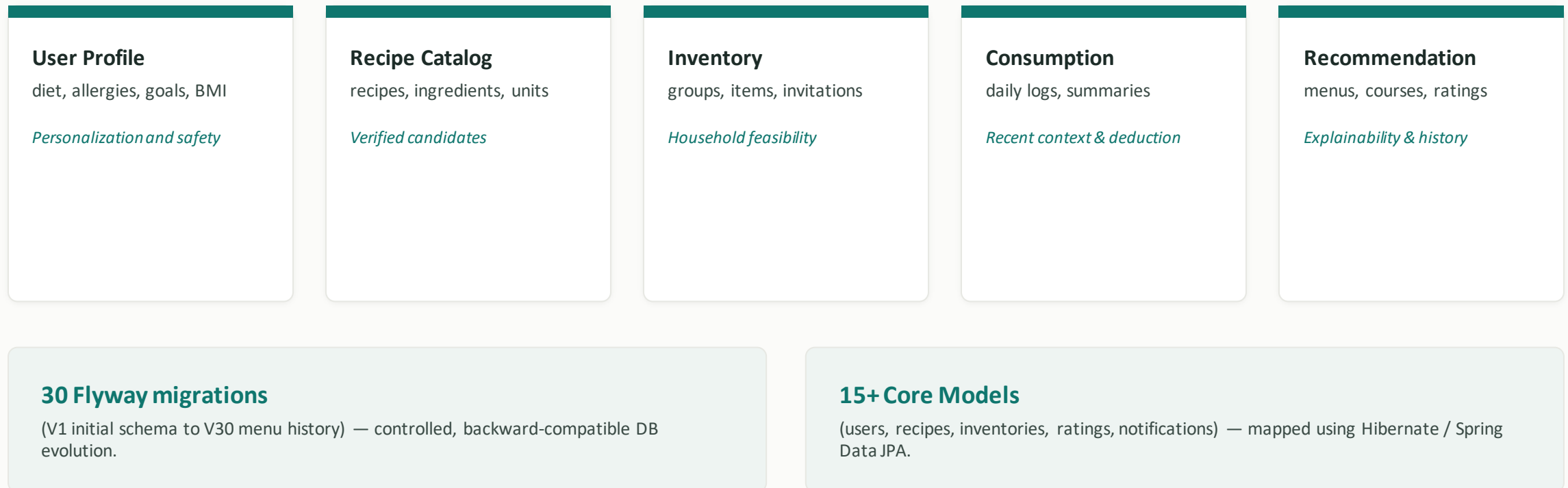
The backend owns the core business logic and is the single source of truth.

Centralized

- Scoring & filtering
- Unit conversion
- Daily logs math
- Stock deduction

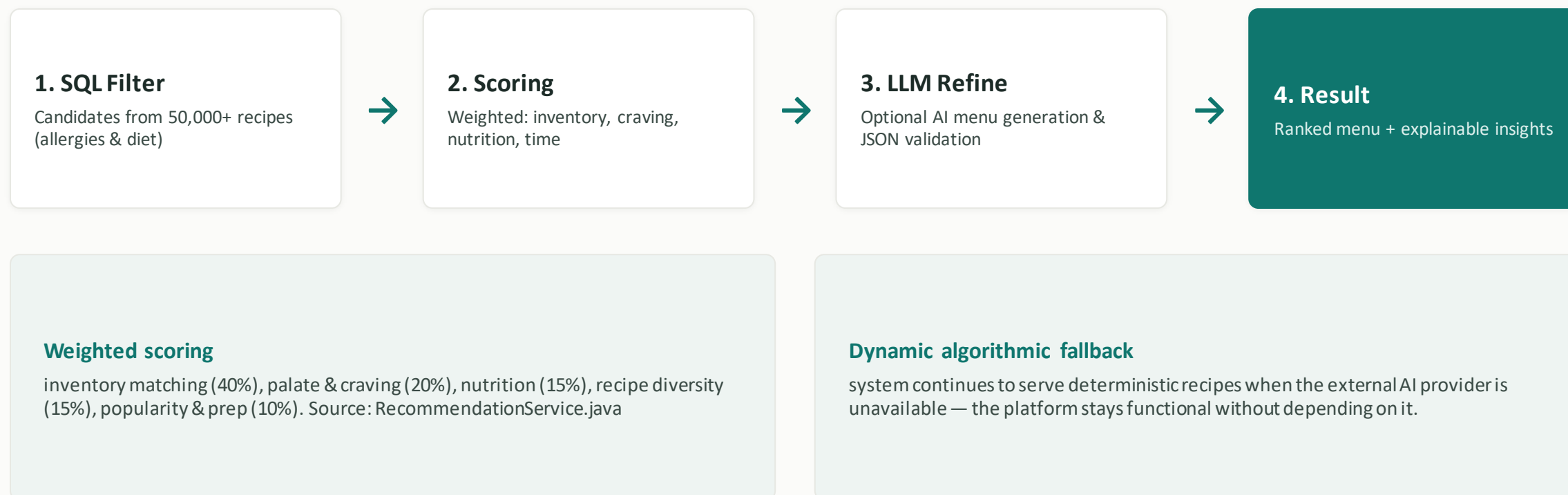
Data & Persistence

The schema evolved from a recipe catalog into a history-aware platform model.



Hybrid Recommendation Engine

AI sits after deterministic filtering and ranking, not before it.



Responsible AI Design

The system reduces hallucination risk by constraining what AI is allowed to influence.

Provider abstraction

OpenAI, Gemini, Claude, and generic endpoints share an AiProvider contract with dynamic Spring Retry and Fallback Chain.

Validation boundary

AI receives only curated candidate pools and JSON responses are parsed, sanitized and validated before use.

Fallback behavior

FREE mode and provider failures still return deterministic recommendations with generated insights.

Hallucination risk is reduced through backend filtering, deterministic scoring, structured response validation and fallback.
AI is not the authority.

Inventory & Consumption

Real-world usefulness comes from shared stock, consumption logging and unit-aware deduction.

Shared Inventory

groups + members sharing stock with invitation controls

Invitation System

accept / reject / cancel invitations between inventories

Stock Action

add / set / subtract with localized unit systems

Prepare Recipe

recipe to portion consumption with automatic stock deduction

Daily Log

track calories, protein, carbs and fats in database

Shopping List

triggers low/missing stock alerts in group view

Frontend Experience

The frontend is workflow-oriented: each feature has its own module but shares auth, API and UI infrastructure.

Dashboard Page

daily calorie logs & pantry indicator progress ring

Recommendations

provider selector, cravings inputs, saved menu history

Recipes Module

detailed search, submissions, custom portions modal

Inventory Module

group view, custom item add, invitations center

Consumption Logging

calories & macro trackers, manual/automatic logs

Profile & Preferences

diet goals, allergen exclusions, negative-cravings

Shared infrastructure: AuthContext + KeycloakAuthService, ServiceRegistry DI, Axios interceptor, i18next EN/TR localization.

Security & Runtime

Reliable operation depends on reproducible runtime boundaries and backend-enforced authorization.

Authentication

Keycloak OIDC manages login, tokens, realm roles and frontend session flow.

Authorization

Spring Security validates JWTs; /api/v1/admin/** requires ADMIN role; backend checks ownership/group membership.

Runtime

Docker Compose starts PostgreSQL, Keycloak, MinIO, backend and frontend with infra/full profiles.

3030

Frontend SPA

8081

Backend API

8080

Keycloak Auth

5432

PostgreSQL

Testing & Quality

Backend quality is covered across layers; frontend automation remains an acknowledged future gap.

47

backend test files (*Test.java)

260

@Test methods verified

Testcontainers integrations

- Real PostgreSQL persistence checks
- Real MinIO image uploads
- Keycloak context security tests

Layer Verifications In Action

Domain Layer

- recommendation scoring logic, recipe allergen filters, inventory rules

Application Layer

- controllers, DTO mappings, exception boundaries, Spring Security roles

Infrastructure Layer

- MinIO adapter, Flyway DB migration seeding, dynamic AI providers

Known gap: broader frontend component and E2E browser automation coverage remains a roadmap item.

Live Demo Flow

Now, live: one end-to-end loop from login to logged consumption.

1. Login

Keycloak OIDC login
with credentials

2. Profile

Setup diet goals,
allergies and macros

3. Pantry

Invite group and seed
available items

4. Recommend

Ask AI with cravings /
fallback mode

5. Prepare

Portion deduction from
shared pantry

6. Logs

Confirm calories and
history update

If external AI is unavailable, FREE mode runs the same flow with deterministic fallback

Sustainable Development Goals

Aligning personalized meal recommendation with global sustainability goals.

SDG 2

Zero Hunger

Better nutrition & meal planning.

SDG 3

Good Health & Well-being

Personalized, goal-driven diets.

SDG 12

Responsible Consumption

Inventory tracking & low-stock alerts help reduce food waste.

Beyond an academic project, Me-AI aligns personalized nutrition with global sustainability goals.

Results & Future Work

The final outcome is a controlled AI-assisted meal platform with clear engineering limits and a practical next roadmap.

Delivered Achievements

- ▶ Secure modular Spring Boot backend and feature-rich React SPA
- ▶ Shared group inventory, shopping list alerts, consumption logger
- ▶ Multi-provider hybrid AI recommendation engine with automatic retry
- ▶ Dockerized runtime (Vite, API, Keycloak, PostgreSQL, MinIO)

Roadmap & Limitations

- ▶ Nutrition estimates are not medical advice; household unit converter is approximate; Compose is not production hardening
- ▶ Future: frontend component tests, TLS production secrets, observability integrations, preference profiling, rich analytical panels

Thank you for listening.

GitHub: github.com/CankayaUniversity/ceng-407-408-2025-2026-AI-Powered-Personalized-Meal-Recommendation-Platform